

3.3 SYSTEM EMERGENCY/MALFUNCTION CONDITIONS

3.3.1 Electrical Short Circuit

Conditions/Indications

- Voltmeter indicates low voltage
- Ammeter may indicate excessive current
- Failure of equipment powered by affected bus
- **GEN 1** and/or **GEN 2** warning light may come on

Procedure

1. Both **GENERATOR** switches – **OFF**
2. **CURRENT IND** switch – **BUS BAR**
3. Electrical consumption – **Reduce**
4. Ammeter and voltmeter – **Monitor**
5. **LAND AS SOON AS PRACTICABLE**

Residual Battery Endurance					
Continuous load [A]	15	20	25	30	40
Time [min]	60	45	35	30	22
NOTE Calculations are based on an assumed minimum battery capacity of 15 Ah. Times include 10-min landing light operation and 10-min radio transmitting.					

If electrical short circuit indications continue:

6. **Battery switch** – **Off if flight conditions permit**

7. **LAND AS SOON AS POSSIBLE**

4. NORMAL PROCEDURES

4.1 STARTING ENGINES

NOTE • At ambient temperatures below +4 °C, refer to “Cold Weather Starting Information” in the basic Flight Manual, Section 4.

NOTE (Continued)

- Either engine may be started first.
- If, for any reason, a starting attempt is discontinued the entire starting sequence must be repeated from the beginning.
- When a starting attempt is aborted but TOT limits were not exceeded, wait 15 seconds after N_1 has returned to zero before attempting restart to permit excess fuel to drain from combustion chamber.
- If a start is not successful after three attempts, maintenance action is required.
- Before starting engines, the operational hydraulic system shall always be pre-selected to system 2 (collective lever lock engaged).

Fire guard (if available)	– Posted
Rotor area	– Clear
Anti-collision light	– On

4.1.1 Starting First Engine

NOTE Operate only the controls of the corresponding engine.

CAUTION IMMEDIATELY ABORT START FOR ANY OF THE FOLLOWING:

- MAIN ROTOR IS NOT ROTATING AT 25% N_1 . (A SECOND AND THIRD START ATTEMPT MAY BE MADE. IF POWER TURBINE STILL FAILS TO ROTATE, DISCONTINUE ATTEMPT).
- STAGNATED START (FUEL STARVATION).
- TOT RISES ABNORMALLY AND RAPIDLY APPROACHES START LIMIT (927 °C).
- TOT EXCEEDS 10-SECOND LIMIT WITHIN TEMPERATURE RANGE (811-927 °C).
- NO POSITIVE ENGINE OR TRANSMISSION OIL PRESSURE INDICATIONS UPON REACHING IDLE SPEED (N_1 59-65%).
- IDLE SPEED IS NOT REACHED WITHIN 60 SECONDS UNLESS N_1 AND N_2 ARE ACCELERATING AND TOT IS WITHIN LIMITS.
- N_2 INCREASES BEYOND N_{Ro} . A FREEWHEELING UNIT SLIPPAGE IS SUSPECTED AND MAINTENANCE ACTION IS REQUIRED.
- AUTO-ACCELERATION (BENDIX FUEL CONTROL SYSTEM INSTALLED). SEE “COLD WEATHER STARTING INFORMATION” IN BASIC FLM, SECTION 4.
- EITHER **HYD 1** OR **HYD 2** WARNING LIGHT REMAINS ON UPON REACHING GROUND IDLE SPEED (60% N_{Ro}).
- **START** WARNING LIGHT REMAINS ON AFTER PLACING RESPECTIVE STARTER/IGN SWITCH TO OFF.

ABORT START PROCEDURE

- POWER LEVER – OFF
- FUEL SUPPLY PUMPS – OFF
- STARTER/IGN SWITCH – VENT, AS NECESSARY TO LOWER TOT
- STARTER/IGN SWITCH – OFF

TOT – Less than 150 °C, otherwise ventilate engine

Fuel supply pump – On

STARTER/IGN switch – ON; start clock; **START** warning light on

N₁ – Check increasing

CAUTION POWER LEVER MUST NOT BE MOVED FROM OFF POSITION UNTIL REQUIRED N₁ HAS BEEN REACHED. STARTING AT N₁ SPEEDS LESS THAN 12% INCREASES THE POSSIBILITY OF EXCEEDING ENGINE TOT LIMITS.

As N₁ accelerates through 12-15%:

If CECO fuel control installed:

Power lever – Slowly advance to a position slightly before IDLE (trim start)

If BENDIX fuel control installed:

Power lever – Move direct to a position slightly before IDLE

NOTE Do not wait for N₁ to peak out. Move power lever towards IDLE immediately upon reaching the desired N₁ speed. Delay in moving power lever may diminish battery capacity early in the start cycle.

TOT and N₁ – Monitor closely, observe TOT peak

Power lever – Lock in IDLE

N₂/N_{Ro} – Check increasing

Engine oil pressure indicators – Check positive indication

At 58% N₁:

STARTER/IGN switch – OFF; **START** warning light off